

CONTACT INFORMATION

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RESEARCH INTERESTS

My Ph.D. research has centered on Applied Cryptography, Computer Security and Privacy, with a primary focus on Searchable Encryption, Multiparty Computation, Oblivious RAM, and Verifiable Computation. I am also interested in designing hardware accelerators using FPGA and GPU technologies, particularly for applications in Cryptography and AI/ML.

EDUCATION

Virginia Tech, Blacksburg, Virginia, United States Aug 2021 – May 2026 (Expected)

Ph.D. Student, Computer Science (M.Sc. Degree awarded in May 2025)

- Dissertation: Trustworthy, Privacy-Preserving and Functional Data Outsourcing Systems
- Advisor: Dr. Thang Hoang
- GPA: 3.96/4

Hanoi University of Science and Technology (HUST), Hanoi, Vietnam Aug 2014 – June 2019

Engineer's Degree (equivalent to B.Sc. Degree), School of Information and Communications Technology (SoICT)

- Thesis: *Research and Implementing Fusion Algorithms on FPGA*
- GPA: 3.83/4 – Valedictorian Graduation Honor (Ranked 1st out of 626)

PROFESSIONAL EXPERIENCES

Graduate Research Assistant, Computer Science, Virginia Tech, Blacksburg, VA Aug 2021 – current

Graduate Teaching Assistant, Computer Science, Virginia Tech, Blacksburg, VA Aug 2021 – current

Applied Scientist Intern, Amazon Development Center U.S., Inc., Seattle, WA Sep 2025 – Dec 2025

Ph.D. SWE Intern, Meta Platforms, Inc., Menlo Park, CA May 2025 – Aug 2025

FPGA Engineer, Viettel High Tech Industries Corp., Hanoi, Vietnam Aug 2019 – July 2021

FPGA Engineer Intern, Viettel High Tech Industries Corp., Hanoi, Vietnam July 2018 – June 2019

Embedded Engineer Intern, Viettel High Tech Industries Corp., Hanoi, Vietnam July 2017 – Dec 2017

PUBLICATIONS

- [1] **Tung Le** and Thang Hoang, "Hermes: Efficient and Secure Multi-Writer Encrypted Database", in *The 46th IEEE Symposium on Security and Privacy (Oakland 2025)*, May 2025, San Francisco, CA, USA. (Acceptance rate: 14.8%)
doi:10.1109/SP61157.2025.00184
- [2] **Tung Le**, Rouzbeh Behnia, Jorge Guajardo, and Thang Hoang, "MUSES: Efficient Multi-User Searchable Encrypted Database", in *The 33rd USENIX Security Symposium (USENIX Security 2024)*, August 2024, Philadelphia, PA, USA. (Acceptance rate: 18.3%)

- [3] **Tung Le**, Pengzhi Huang, Attila A. Yavuz, Elaine Shi, and Thang Hoang, “Efficient Dynamic Proof of Retrievability for Cold Storage”, in *The 30th Annual Network and Distributed System Security Symposium (NDSS 2023)*, February 2023, San Diego, CA, USA. (Acceptance rate: 16.2%)
doi:10.14722/ndss.2023.23307
- [4] **Tung Le** and Thang Hoang, “MAPLE: A Metadata-Hiding Policy-Controllable Encrypted Search Platform with Minimal Trust”, in *The 23rd Privacy Enhancing Technologies Symposium (PETS 2023)*, July 2023, Lausanne, Switzerland. (Acceptance rate: 21.8%)
doi:10.56553/popets-2023-0105
- [5] **Le Thanh Tung**, Le Thanh Bang, Pham Trong Thuy, Nguyen Duc Hoan, and Vuong Dang Huy, “Implement Detail Enhancement Algorithm on FPGA for Real-Time and Energy-Efficient Embedded Systems”, in *The 8th IEEE International Conference on Communications and Electronics (IEEE ICCE 2020)*, January 2021, Phu Quoc, Kien Giang, Vietnam. (Acceptance rate: 49.7%)
doi:10.1109/ICCE48956.2021.9352095

AWARDS AND HONORS

- 2023 CCI SWVA Cyber Innovation Scholar, Virginia Tech, USA 2023
- Advanced Labor Performance, Viettel High Tech Industries Corp., Hanoi, Vietnam 2020
- SoICT Valedictorian Graduation Honor, HUST, Hanoi, Vietnam 2019
- Best Presentation Award in Thesis Defense, SoICT, HUST, Hanoi, Vietnam 2019
- HUST Excellence Scholarship Award for AY 2018–2019, HUST, Hanoi, Vietnam 2018
- HUST Excellence Scholarship Award for AY 2017–2018, HUST, Hanoi, Vietnam 2017
- Excellent Student Award in AY 15-16, HUST, Hanoi, Vietnam (Top 20 out of 25k students) 2016
- Scholarships for Excellent Academic Record, HUST, Hanoi, Vietnam 2014–2016

TEACHING EXPERIENCE

I have instructed students and graded assignments and exams in the following courses at Virginia Tech:

- CS5594 - Blockchain Technologies (Graduate Level): Spring 2024 (53 students)
- CS5584 - Network Security (Graduate Level): Fall 2023 (46 students)
- CS1044 - Introduction to Programming in C (Undergraduate Level): Spring 2022 (69 students)
- CS3114 - Data Structures and Algorithms (Undergraduate Level): Fall 2021 (414 students), Fall 2024 (481 students)

PROFESSIONAL SERVICES

Conference Reviewer: ACSAC (2022–2024), PETS (2024), USENIX (2025)

Journal Reviewer: IEEE Transactions on Information Forensics and Security (2024-2025), IEEE Transactions on Dependable and Secure Computing (2025)

TALKS

Derived-Key Mode Primitive in AWS-LC

- Presented at Amazon Web Services Seattle, WA, USA, Dec 2025

Hermes: Efficient and Secure Multi-Writer Encrypted Database

- Presented at IEEE S&P (Oakland) 2025 San Francisco, CA, USA, May 2025

Trustworthy, Privacy-Preserving and Functional Data Outsourcing Systems

- Ph.D. Preliminary Exam, Virginia Tech Blacksburg, VA, USA, Dec 2024

MUSES: Efficient Multi-User Searchable Encrypted Database

- Presented at USENIX Security 2024 Philadelphia, PA, USA, Aug 2024

MAPLE: A Metadata-Hiding Policy-Controllable Encrypted Search Platform with Minimal Trust

- Presented at PETS 2023 Remote, July 2023

Poster: Efficient Dynamic Proof of Retrievability for Cold Storage

- Presented at CCI Student Researcher Showcase 2023 Blacksburg, VA, USA, Apr 2023

Efficient Dynamic Proof of Retrievability for Cold Storage

- Presented at ISOC NDSS 2023 San Diego, CA, USA, Mar 2023

Implement Detail Enhancement Algorithm on FPGA for Real-Time and Energy-Efficient Embedded Systems

- Presented at IEEE ICCE 2020 Phu Quoc, Kien Giang, Vietnam, Jan 2021

RESEARCH EXPERIENCE

Graduate Research Assistant

Aug 2021–Present

Department of Computer Science, Virginia Tech, Blacksburg, VA

- Develop algorithms and implement searchable encrypted systems using C/C++ including:
 - **Hermes**: a multi-writer searchable encrypted database that can prevent keyword-guessing attacks, achieve optimal search complexity (sublinear in keyword set size) and forward privacy with user efficiency. My developed source code is publicly available at <https://github.com/vt-asaplab/Hermes>.
 - **MUSES**: a multi-server searchable encrypted system with multi-writer support. It hides all statistical information including search, result and volume patterns while achieving minimal user overhead for keyword search and key rotation. My artifact has been awarded all three badges (available, functional, reproducible) at USENIX Security 2024: <https://secartifacts.github.io/usenixsec2024/results>.
 - **MAPLE**: a multi-server searchable encryption design attaining a high level of security. It protects search, result and volume patterns under malicious adversaries, supports multi-user with fine-grained access control, also optimizes server-side computational complexity. My developed library is publicly available at <https://github.com/vt-asaplab/MAPLE>.
- Develop and implement **Porla** using MATLAB, C/C++ and Golang. Porla is a proof of retrievability protocol that minimizes audit cost by leveraging error-correcting code and verifiable computation techniques, including Bulletproofs and KZG. My developed library is publicly available at <https://github.com/vt-asaplab/porla>.

Applied Scientist Intern

Sep 2025 – Dec 2025

Amazon Development Center U.S., Inc. Seattle, WA

- Perform a comprehensive security analysis of AES-GCM across diverse threat models and against various attacks; analyze the advantages and limitations in terms of security guarantees, as well as computational cost of state-of-the-art derived-key modes for AES-GCM.
- Implement and evaluate the XAES-256-GCM derived-key mode in the AWS-LC library using C/C++, optimize performance using AVX-512 vectorization, and expose APIs for binding libraries Java and Rust.

Ph.D. Software Engineer Intern

May 2025 – Aug 2025

Meta Platforms, Inc. Menlo Park, CA

- Implement, propose and evaluate new approaches/strategies to enhance user engagement and optimize capacity costs in the Instagram Stories Delivery system.
- Stack: Python (Django), Hack, SQL, etc.

FPGA Engineer

Aug 2019–July 2021

Viettel High Technology Industries Corporation, Hanoi, Vietnam

- Develop, integrate and write testbenches for IP cores of 5G Physical layer: implement IP core for resource mapping of PDCCH; integrate Xilinx Polar encoder to accelerate bit processing of PBCH and PDCCH;

optimize, test and debug the overall 5G downlink system including PBCH, PDCCH, PDSCH and CSI-RS on boards VCU-118 (Xilinx) and 21DR (Viettel) (achieved download speed ≈ 1 Gbps and supported 8 concurrent users per time slot); test and debug IP cores and the overall system using Xilinx's tools: Vivado Simulator, Virtual Input/Output (VIO) and Integrated Logic Analyzer (ILA).

- Develop IP core for Digital Detail Enhancement algorithm to enhance the quality of infrared images, including multiple stages of convolutions and computations based on the histogram of the input image; test and debug on Xilinx Zedboard/Zybo Z7.
- Develop IP core interfacing with Quad SPI Flash module to read data from flash memory and write it to DRAM directly, which can speed up the system booting process by approximately 5 $\times$ (delay decreases from 42s to 8s); test and debug on board Neso Artix 7.
- Adapt Linux kernel modules and C libraries from Intel x86 architecture to ARMv8 platform of NXP.

Research Student

Sep 2018–July 2019

Computer System Lab, SoICT, HUST, Hanoi, Vietnam

- Implement **Fusion Algorithms** on **FPGA**, test and debug on boards **DE2i-150** and **DE1-SoC**.
- Develop **Linux device driver** on Ubuntu 14.04 for board DE2i-150 to transfer data between FPGA and DRAM memory (Intel Atom Processor) via PCIe bus using DMA/SG-DMA.
- Develop IP core to transfer data from GNSS receiver circuit to PC via EZ-USB FX2LP module.
- Design PCB using Altium Designer and program microcontrollers, including 8051, PIC (PIC16F84, PIC16F877), and AVR (ATMEGA8, ATMEGA16).

TECHNICAL SKILLS

- **Programming Languages:** C/C++, Python, MATLAB, Golang, Verilog, Git, Bash, LaTeX
- **Embedded System Development:** Microcontroller programming (PIC, AVR), Digital design
- **Operating System/Software:** Linux (Ubuntu/CentOS), Vivado, HLS, System Generator, Tcl script, Quartus, Altera DSP Builder, Simulink, GDB
- **Algorithm Design:** Searchable Encryption, Verifiable Computation, Multiparty Computation
- **Network Programming and Multithreading/IPC:** TCP/UDP, Mutex/Semaphore, Shared Memory/Message Queue (POSIX and System V)